

671-billion-parameter AI model DeepSeek-V3. Within a few weeks came DeepSeek-R1, an AI model with chain-of-thought reasoning capabilities, trained using large-scale reinforcement learning with supervised fine-tuning. Chain-of-thought reasoning means that R1 can share the thought process or step-by-step reasoning by which it arrives at an answer. For example, you can see how it solves a math problem step-by-step. It stops and evaluates each step, sometimes even going back to correct mistakes. What is more, it delivers results comparable to OpenAI's ChatGPT o1 in various tasks, including math, coding, reasoning, and creative writing, at a fraction of the cost. This created waves—actually, a tsunami—in the tech world, shaking up not just the AI industry but the stock market as well!

DeepSeek claims to have trained its 671-billion-parameter AI model in less than two months using around 2000 GPUs (including Nvidia H800 chips that Wenfeng had purchased before the embargo), at a cost lower than \$6 million. Other AI majors have reported using approximately 15,000 GPUs, and experts have pegged the cost of training traditional models to be between \$100 million and \$1 billion. Some independent research firms have raised questions on these claims—claiming that DeepSeek might have somehow worked around the US embargo and procured and used more GPUs or that they might not have disclosed indirect costs like that of experimentation and research, but most agree that its resource usage and cost are significantly lower than traditional models.

“WITHIN THE DAILY FLOW OF INNOVATION IN THE US, WHAT WE DID IS ACTUALLY QUITE ORDINARY. THEY’RE SURPRISED BECAUSE WE’RE A CHINESE COMPANY STEPPING INTO THEIR DOMAIN AS AN INNOVATOR, WHEREAS MOST CHINESE COMPANIES ARE USED TO SIMPLY FOLLOWING.”



— LIANG WENFENG, CEO OF DEEPSEEK
(TRANSLATED FROM AN INTERVIEW IN CHINESE)

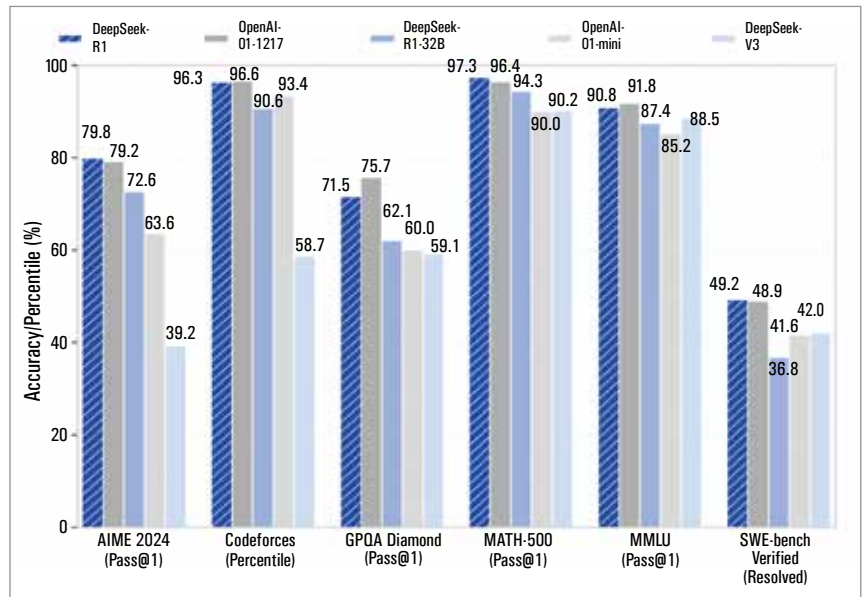


Figure 1: DeepSeek-R1 performs on par with ChatGPT o1 (Source: DeepSeek API Docs)

How did they reduce the costs?

“The fact that DeepSeek could be built for less money, less computation, and less time, and can be run locally on less expensive machines argues that as everyone was racing towards bigger and bigger, we missed the opportunity to build smarter and smaller,” said Kristian Hammond, professor of computer science at Northwestern University, in a story in *Live Science*. Indeed, while huge corporate entities booked their H100s from Nvidia and waited patiently for it to be delivered, elsewhere in the world, someone who did not have access to it used the opportunity to come up with an AI model that could run on less powerful processors!

Note that the model size is not small. Yet, how does it run with lower

resource requirements? Here are some of the tricks that make DeepSeek tick!

- **Mixture-of-Experts (MoE):** DeepSeek’s models are split into sub-models, each specialising in a specific task or data type. When there is a query from the user, a router transfers it to the relevant expert / sub-model. Of course, some jack-of-all-trades sub-models handle queries that cannot be properly classified and routed. According to a technical report published by the DeepSeek team, although the DeepSeek-V3 model has 671 billion parameters, only 37 billion are activated for any given token.
- **Inference-time compute scaling:** DeepSeek uses a load-balancing technique that dynamically scales computing up or down to match the complexity of an assigned task.
- **Mixed-precision framework:** In AI, weights are numbers determining the strength and direction of connections between neurons in artificial neural networks. They determine the model’s ability to learn and make predictions. Instead of using 32-bit floating point numbers (FP32) for all the